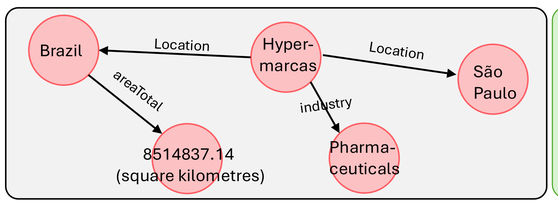
What Gets Unmasked First? Trajectory Analysis of Diffusion Models for Graph-to-Text Generation

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Motivation



Graph to Text Model Generated Sentence

Hypermarcas is a pharmaceutical company located in São Paulo, Brazil. Hypermarcas is located in Brazil, which has a total area of 8514837.14 square kilometres. Brazil has a total area of 8514837.14 square kilometres.

Hallucination

Autoregressive large language models

The						
The	neutral					
The	neutral	point				
The	neutral	point	of			
The	neutral	point	of	view		
The	neutral	point	of	view	policy	...

LLaMA 3.1, LLaMA 3.3 , Qwen 3

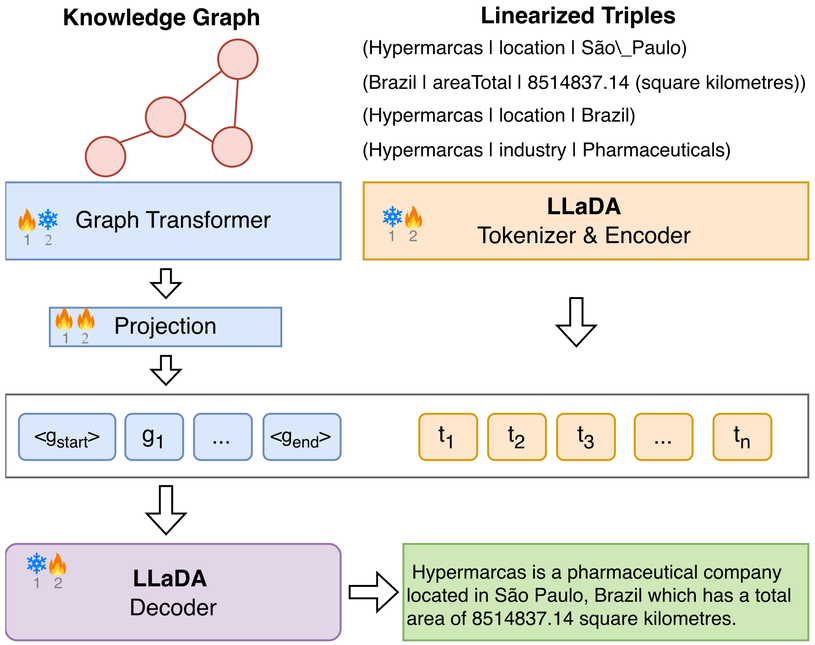
Masked diffusion models (MDLMs)

The	[MASK]	[MASK]	[MASK]	[MASK]	[MASK]	[MASK]
The	[MASK]	[MASK]	[MASK]	[MASK]	policy	[MASK]
The	[MASK]	[MASK]	[MASK]	view	policy	[MASK]
The	neutral	[MASK]	[MASK]	view	policy	[MASK]
The	neutral	point	[MASK]	view	policy	[MASK]
The	neutral	point	of	view	policy	...

LLaDA-8B, Dream-7B

- Fixed left-to-right order
- One token at a time
- Earlier decisions cannot be revised
- Flexible, confidence-based unmasking
- Unclear unmasking pattern
- Unclear performance of MDLMs

Graph-LLaDA



Stage 1
- Freeze LLaDA
- Train Graph Transformer + Projector
- Align graph representations with LLaDA's embedding space

Stage 2
- Freeze Graph Transformer
- Train Projector + LoRA adapters
- Adapt LLaDA to graph-conditioned inputs

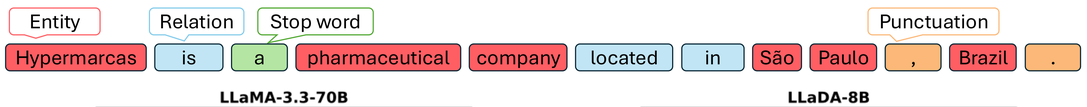
λ	BLEU-4	METEOR	ROUGE_L	CIDEr
Graph-LLaDA-8B (Stage 1)				
1.00	31.5	28.7	57.8	2.22
0.50	49.1	37.6	65.5	2.86
0.25	48.8	42.3	67.2	3.09
Graph-LLaDA-8B (Stage 1 + Stage 2)				
1.00	36.3	32.8	64.6	3.18
0.50	51.8	39.5	67.7	3.03
0.25	55.4	43.4	69.5	3.37

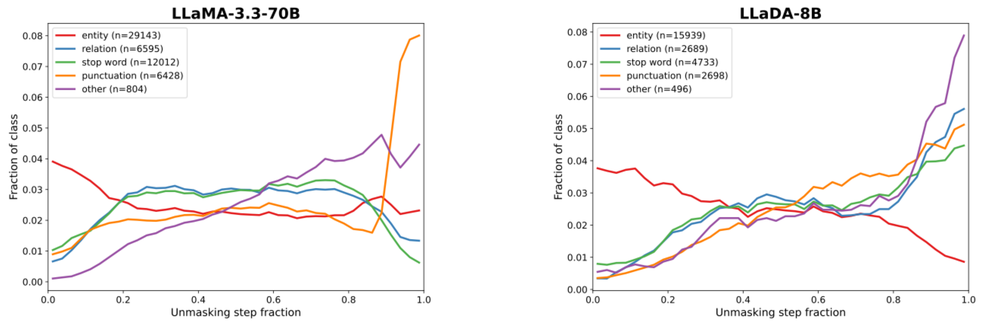
Graph-LLaDA combines graph structure with diffusion decoding and shows strong cross-dataset generalization.

Trajectory

Trajectory recording: For each committed token, record its output position and commitment timestep. Normalize the timestep to $[0, 1]$ to enable comparison across sequence lengths and different denoising schedules.

Token classification: Classify each token as **entity, relation, stop word, punctuation,** or **other** using a priority-based matching procedure.





MDLMs exhibit strong entity-first generation.

λ – Scaled Structural Decoding

Matrix	LLaDA	LLaDA SFT
Content Before "."	60.9%	17.8%
Mean First "."	0.652	0.144
Median First "."	0.835	0.062
Periods / Sample	1.22	2.44
EOS / Sample	37.2	21.4

λ	BLEU-4	METEOR	ROUGE_L	CIDEr
LLaDA-8B SFT				
1.00	44.3	41.5	63.5	2.17
0.75	45.6	41.8	65.1	2.38
0.50	53.7	43.4	68.6	3.18
0.25	45.0	41.8	65.5	2.41

$$\tilde{c}_i = \begin{cases} \lambda c_i, & y_i \in \{\text{punctuation, EOS, EOT}\} \\ c_i, & \text{Otherwise} \end{cases}$$

λ – Scaled structural decoding mitigates premature commitment.

Evaluation

Metrics	Human Evaluation	LLM-Judge
• BLEU-4 / METEOR / ROUGE_L / CIDEr • Reference-based Lexical overlap • Limited for factual error	• 50 WebNLG samples • 2 human annotators • Fluency / Hallucination / Omission	• 1,000 pairwise comparisons • Two judges + reversed output order • Elo ratings for fluency / hallucination / omission

Results

Model	BLEU-4	METEOR	ROUGE_L	CIDEr
Supervised Baselines				
GAP (LAGRANGE)‡	23.3	28.9	40.8	1.78
MGSA†	**66.5**	46.9	**76.5**	–
GAP (WEBNLG)‡	66.3	46.9	76.3	**4.65**
JointGT†	65.9	**47.2**	76.1	–
Autoregressive Baselines				
LLaMA 3.1-8B ZS	44.5	37.9	61.2	2.64
Qwen3-8B ZS	48.4	40.5	63.8	2.90
LLaMA 3.3-70B ZS	51.4	41.0	66.9	3.15
LLaMA 3.1-8B SFT	57.2	42.6	68.8	3.38
LLaMA 3.3-70B SFT	**57.6**	**42.8**	**69.4**	**3.44**
Masked Diffusion Models				
Dream-7B ZS	23.4	25.5	56.9	2.23
LLaDA-8B ZS	42.2	36.1	62.3	2.50
LLaDA-8B SFT	44.3	41.5	63.5	2.17
LLaDA-8B SFT, λ=0.5	53.7	**43.4**	68.6	3.18
Graph-LLaDA λ=0.25	**55.2**±0.4	43.3±0.3	**69.1**±0.8	**3.36**±0.03

LLaDA-8B: Hypermarcas is a pharmaceutical company located in São Paulo, Brazil.

Omission

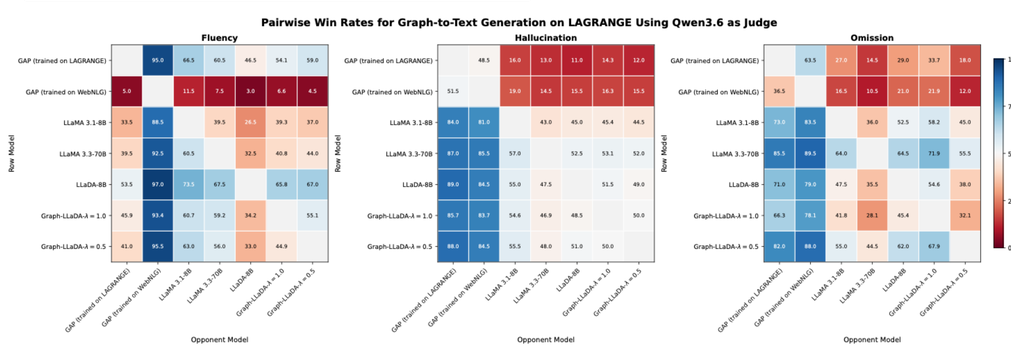
LLaDA-8B SFT ($\lambda = 0.5$): Hypermarcas is a pharmaceutical company located in São Paulo, Brazil. Hypermarcas is located in Brazil, which has a total area of 8514837.14 square kilometres. Brazil has a total area of 8514837.14 square kilometres.

Hallucination

Graph-LLaDA ($\lambda = 1.0$): Hypermarcas is a pharmaceutical company located in São Paulo, Brazil.

Omission

Graph-LLaDA ($\lambda = 0.5$): Hypermarcas is a pharmaceutical company located in São Paulo, Brazil which has a total area of 8514837.14 square kilometres.



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